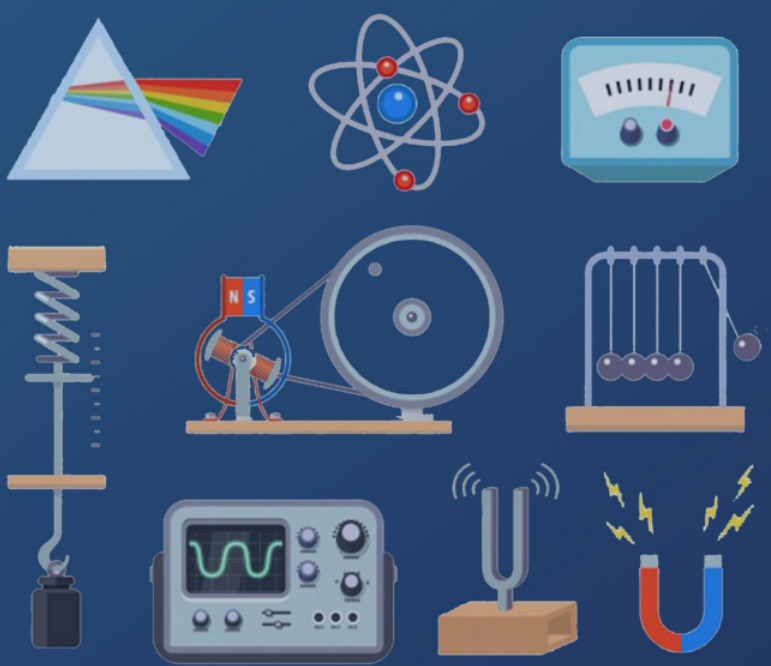
PHYSICS

Gravitation



SHEET

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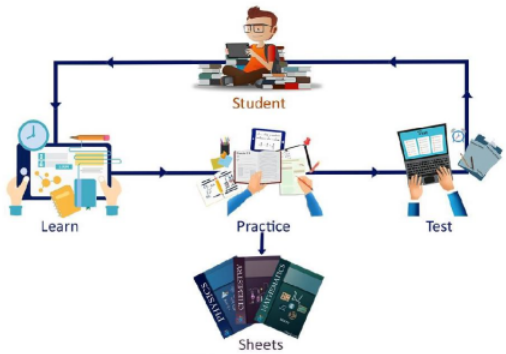
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex 1	Ex 1A	Ex 2	Ex 2A
1	Universal Law of Gravitation	1-3	14	1,2	1
2	Gravitational Field	4-6	1	3	17
3	Acceleration due to Gravity	7-9	13	4,5	2,3,18
4	Gravitational Potential Energy and Potential	10-12	2-4,11	6	
5	Motion of Satellite	13-20	5,6,12,15	7-9	4-6,12-14
6	Kepler's Laws	21-23	7,9,10,16,17	10-12	7,15,16
7	Escape Velocity	24-29		13,14	8,9
8	Weightlessness	30		15	10
9	After full chapter		8	16	11,19,20

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Universal Law of Gravitation

Q1. The mass of the moon is 1% of mass of the earth. The ratio of gravitational pull of earth on moon to that of moon on earth will be:-

(A) 1 : 1

(B) 1 : 10

(C) 1 : 100

(D) 2 : 1

Q2. Three identical point masses, each of mass 1 kg lie in the x–y plane at points (0, 0) (0, 0.2m) and (0.2m,0). The gravitational force on the mass at the origin is :–

(A) $1.67 \times 10^{-11}\ (\hat{i}+\hat{j})N$

(B) $3.34 \times 10^{-10}\ (\hat{i}+\hat{j})N$

(C) $1.67 \times 10^{-9}\ (\hat{i}+\hat{j})N$

(D) $3.34 \times 10^{-10}\ (\hat{i}-\hat{j})N$

Q3. Four particles of masses m, 2m, 3m and 4m are kept in sequence at the corners of a square of side a. The magnitude of gravitational force acting on a particle of mass m placed at the centre of the square will be :

(A) $\frac{24m^2G}{a^2}$ (B) $\frac{6m^2G}{a^2}$

(C) $\frac{4\sqrt{2}Gm^2}{a^2}$ (D) Zero

Gravitational Field

Q4. The radius of Earth is about 6400 km and that of mars is 3200 km. The mass of the Earth is 10 times the mass of mars. An object weight 200 N on the surface of Earth. Its weight on the surface of mars will be :–

(A) 80 N (B) 40 N (C) 20 N (D) 8 N

Q5. The acceleration due to gravity g and mean density of earth ρ are related by which of the following relations? [G = gravitational constant and R = radius of earth] :

(A) $\rho=\frac{4\pi gR^2}{3G}$ (B) $\rho=\frac{4\pi gR^3}{3G}$

(C) $\rho=\frac{3g}{4\pi GR}$ (D) $\rho=\frac{3g}{4\pi GR^3}$

Q6. If the radius of the earth be increased by a factor of 5, by what factor its density be changed to keep the value of g the same?

(A) 1/25 (B) 1/5 (C) $1/\sqrt{5}$ (D) 5

Acceleration due to Gravity

Q7. A stone drop from height 'h' reaches to earth surface in1 sec. If the same stone taken to moon and drop freely then it will reaches from the surface of the moon in the time :–

(A) $\sqrt{6}$ sec (B) 9 sec

(C) $\sqrt{3}$ sec (D) 6 sec

Q8. Consider the earth to be a homogenous sphere, scientist A goes deep down in a mine and scientist B goes high up in a ballon. The gravitation field measured by :–

(A) A goes a decreasing and the by B goes an increasing

(B) B goes a decreasing and that by A goes a increasing

(C) each remains unchanged

(D) each goes on decreasing

Q9. Gravitation on moon is $1/6^{th}$ of that on earth. When a balloon filled with hydrogen is released on moon then this :-

(A) Will rise with an acceleration less than $\left(\frac{g}{6}\right)$

(B) Will rise with acceleration $\left(\frac{g}{6}\right)$

(C) Will fall down with an acceleration less than $\left(\frac{5g}{6}\right)$

(D) Will fall down with acceleration $\left(\frac{g}{6}\right)$

Gravitational Potential Energy and Potential

Q10. A body attains a height equal to the radius of the earth when projected from earth' surface. The velocity of the body with which it was projected is :–

(A) $\sqrt{\frac{GM}{R}}$ (B) $\sqrt{\frac{2GM}{R}}$

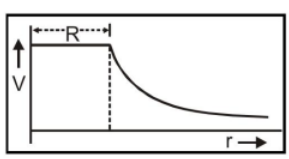
(C) $\sqrt{\frac{5}{4}\frac{GM}{R}}$ (D) $\sqrt{\frac{3GM}{R}}$

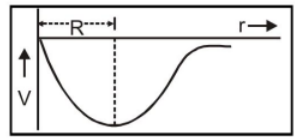
Q11. The gravitational potential energy of a body at a distance r from the center of the earth is U. The gravitational force at that point is :-

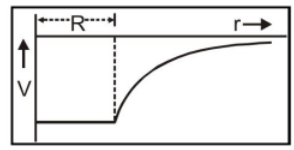
(A) $\frac{U}{r^2}$ (B) $\frac{U}{r}$

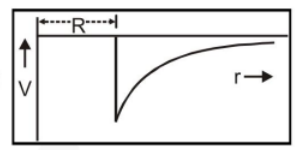
(C) Ur (D) Ur^2

Q12. Which of the following waves expresses the variation of gravitational potential with distance for a hollow sphere of radius R :

(A) 

(B) 

(C) 

(D) 

Motion of Satellite

Q13. What will be velocity of a satellite revolving around the earth at a height h above surface of earth if radius of earth is R :-

(A) $R^2\sqrt{\frac{g}{R+h}}$ (B) $R\frac{g}{(R+h)^2}$

(C) $R\sqrt{\frac{g}{R+h}}$ (D) $R\sqrt{\frac{R+h}{g}}$

Q14. Two identical satellites are at the heights R and 7R from the earth's surface. Then which of the following statement is incorrect :–

(R = radius of the earth)

(A) ratio of kinetic energy of both is 4

(B) ratio of total energy of both is 5

(C) ratio of potential energy of both 4

(D) ratio of total energy of both is 4

Q15 Two artificial satellites of masses m_1 and m_2 are moving with speeds v_1 and v_2 in orbits of radii r_1 and r_2 respectively. If $r_1 > r_2$ then which of the following statements is true :–

(A) $v_1 = v_2$ (B) $v_1 > v_2$

(C) $v_1 < v_2$ (D) $\frac{v_1}{r_1} = \frac{v_2}{r_2}$

Q16. Two artificial satellites A and B are at a distance r_A and r_B above the earth's surface. If the radius of earth is R, then the ratio of their speed will be :–

(A) $\left[\frac{r_B+R}{r_A+R}\right]^{1/2}$ (B) $\left[\frac{r_B+R}{r_A+R}\right]^{2}$

(C) $\left[\frac{r_B}{r_A}\right]^{2}$ (D) $\left[\frac{r_B}{r_A}\right]^{1/2}$

Q17. One projectile after deviating from its path starts moving round the earth in a cirular path of radius equal to nine times the radius of earth R. Its time period will be :

(A) $2\pi\sqrt{\frac{R}{g}}$ (B) $27\times 2\pi\sqrt{\frac{R}{g}}$

(C) $\pi\sqrt{\frac{R}{g}}$ (D) $0.8\times 3\pi\sqrt{\frac{R}{g}}$

Q18. A satellite is moving in a circular orbit around earth with a speed v, If its mass is m, then its total energy will be :

(A) $\frac{3}{4}mv^2$ (B) mv^2

(C) $\frac{1}{2}mv^2$ (D) $-\frac{1}{2}mv^2$

Q19. The relay satellite transmits the television programme continuously from one part of the world to another because its :

(A) Period is greater than the period of rotation of the earth about its axis

(B) Period is less than the period of rotation of the earth about its axis

(C) Period is equal to the period of rotation of the earth about its axis

(D) Mass is less than the mass of earth

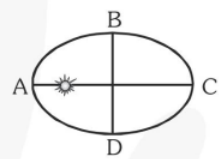
Q20. The gravitational force between two bodies is directly proportional to $\frac{1}{R}$ (not $\frac{1}{R^2}$), where 'R' is the distance between the bodies. Then the orbital speed for this force in circular orbit is proportional to :-

(A) $1/R^2$ (B) R^0

(C) R (D) 1/R

Kepler's Laws

Q21. In adjoining figure earth goes around the sun in elliptical orbit on which point the orbital speed is maximum :-



(A) on A (B) on B

(C) on C (D) on D

Q22. A planet is revolving around the Sun in an elliptical orbit. Its closest distance from the Sun is r_{min}. The farthest distance from the Sun is r_{max}. If the orbital angular velocity of the planet when it is nearest to the Sun is ω, then the orbital angular velocity at the point when it is at the farthest distance from the Sun is–

(A) $\left(\sqrt{\frac{r_{min}}{r_{max}}}\right)\omega$ (B) $\left(\sqrt{\frac{r_{max}}{r_{min}}}\right)\omega$

(C) $\left(\frac{r_{max}}{r_{min}}\right)^2\omega$ (D) $\left(\frac{r_{min}}{r_{max}}\right)^2\omega$

Q23. A planet of mass m is in an elliptical orbit about the sun ($m << M_{sun}$) with an orbital period T. If A be the area of orbit, then its angular momentum would be :

(A) $\frac{2mA}{T}$ (B) mAT

(C) $\frac{mA}{2T}$ (D) 2mAT

Escape Velocity

Q24. Escape velocity of a body from the surface of the earth is 11.2 km/s. Escape velocity, when thrown at an angle of 45° from horizontal will be :–

(A) 11.2 km/s (B) 22.4 km/s

(C) $\frac{11.2}{\sqrt{2}}$ km/s (D) $11.2\sqrt{2}$ km/s

Q25. Escape velocity from earth is 11.2 km/sec. Another planet of same mass has radius $\frac{1}{4}$ times that of earth. What is the escape velocity from this planet ?

(A) 11.2 km/sec (B) 44.8 km/sec

(C) 22.4 km/sec (D) 5.6 km/sec

Q26. A body of mass m is situated at distance $4R_e$ above the Earth's surface, where R_e is the radius of Earth. How much minimum energy be given to the body so that it may escape?

(A) mgR_e (B) $2mgR_e$

(C) $\frac{mgR_e}{5}$ (D) $\frac{mgR_e}{16}$

Q27. The ratio of radii of two satellites is p and the ratio of their acceleration due to gravity is q. The ratio of their escape velocities will be :

(A) $\left(\frac{q}{p}\right)^{1/2}$ (B) $\left(\frac{p}{q}\right)^{1/2}$

(C) pq (D) $\sqrt{pq}$

Q28. A missile is launched with a velocity less than the escape velocity. Sum of its kinetic energy and potential energy is :-

(A) Positive

(B) Negative

(C) May be negative or positive depending upon its initial velocity

(D) Zero

Q29. **Statement – 1 :** Escape velocity is independent of the angle of projection.

Statement – 2 : Escape velocity for vertical projection from the surface of Earth is $\sqrt{\frac{2GM}{R}}$.

(A) Statement-1 is True, Statement–2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement–2 is True; Statement-2 is not a correct explanation for Statement-1.

(C) Statement-1 is True, Statement-2 is False.

(D) Statement-1 is False, Statement-2 is True.

Weightlessness

Q30. **Statement – 1:** An astronaut in an orbiting space station above the Earth experiences weightlessness.

Statement – 2 : An object moving around the Earth under the influence of Earth's gravitational force is in a state of 'free-fall'.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement1

(B) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The magnitudes of the gravitational field at distance r_1 and r_2 from the centre of a uniform, sphere of radius R and mass M are F_1 and F_2 respectively. then :–

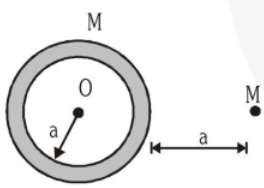
(A) $\frac{F_1}{F_2}=\frac{r_1}{r_2}$ if $r_1 < R$ and $r_2 < R$

(B) $\frac{F_1}{F_2}=\frac{r_2^2}{r_1^2}$ if $r_1 > R$ and $r_2 > R$

(C) $\frac{F_1}{F_2}=\frac{r_1^3}{r_2^3}$ if $r_1 < R$ and $r_2 < R$

(D) $\frac{F_1}{F_2}=\frac{r_1^2}{r_2^2}$ if $r_1 < R$ and $r_2 < R$

Q2. A particle of mass M is at a distance a from surface of a thin spherical shell of equal mass and having radius a :



(A) Gravitational field and potential both are zero at centre of the shell

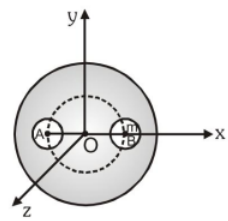
(B) Gravitational field is zero not only inside the shell but at a point outside the shell also

(C) Inside the shell, gravitational field alone is zero

(D) Neither gravitational field nor gravitational potential is zero inside the shell

Q3. A solid sphere of uniform density and radius 4 units is located with its centre at the origin O of coordinates. Two spheres of equal radii 1 unit, with their centres at A (–2, 0, 0) and B (2,0,0) respectively, are taken out of the solid leaving behind spherical cavities as shown in figure.

Then :-



(A) The gravitational field due to this object at the origin is zero

(B) The gravitational field at the point B(2, 0, 0) is zero

(C) The gravitational potential is the same at all points of circle $y^2 + z^2 = 36$

(D) The gravitational potential is the same at all points on the circle $y^2 + z^2 = 4$

Q4. Gravitational potential at the centre of curvature of a hemispherical bowl of radius R and mass M is V :-

(A) Gravitational potential at the centre of curvature of a thin uniform wire of mass M, bent into a semicircle of radius R, is also equal to V

(B) In part (A) if the same wire is bent into a quarter of a circle then also the gravitational potential at the centre of curvature will be V

(C) In part (A) if the wire mass is non uniformly distributed along its length audit is bent into a semicircle of radius R, gravitational potential at the centre is V

(D) None of these

Q5. For a satellite to be geo–stationary, which of the following is/are essential conditions?

(A) It must always be stationed above the equator and it's time period must be 24 hours.

(B) It must be rotated from west to east

(C) It must be about 36000 km above the Earth

(D) Its orbit must be circular, and not elliptical

Q6. The Sun travels in approximately circular orbit of radius R around the center of the galaxy and completes one revolution in time T. The Earth also revolves around the Sun in time t. Assume orbit of the Earth to be a circle of radius r ($r << R$) and whole mass of the galaxy centered on its center. By using only these given informations, find an expression for the ratio of the mass of the galaxy to that of the Sun.

(A) $\left(\frac{R}{r}\right)^3\left(\frac{t}{T}\right)^2$ (B) $\left(\frac{R}{r}\right)^3\left(\frac{T}{t}\right)^2$

(C) $\left(\frac{R}{r}\right)^2\left(\frac{t}{T}\right)^3$ (D) $\left(\frac{R}{r}\right)^2\left(\frac{T}{t}\right)^3$

Q7. When a satellite in a circular orbit around the Earth enters the atmospheric region, it encounters small air resistance to its motion. Then

(A) Its kinetic energy increases

(B) Its kinetic energy decreases

(C) Its angular momentum about the Earth decreases

(D) Its period of revolution around the Earth increases

Q8. A double star is a system of two stars of masses m and 2m, rotating about their centre of mass only under their mutual gravitational attraction. If r is the separation between these two stars then their time period of rotation about their centre of mass will be proportional to :

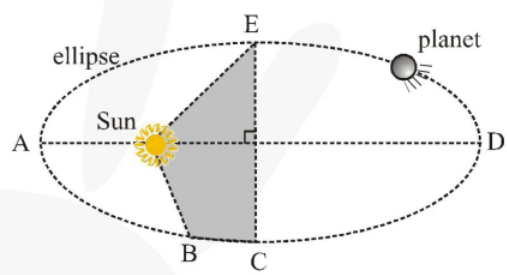
(A) $r^{3/2}$ (B) r (C) $m^{1/2}$ (D) $m^{-1/2}$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 9 & 10

In the picture below, the planet orbits around the sun with a period of 40 months and takes 12 months to translate from the point D to point E and 1 month from point B to C. The area of the ellipse is A. Mass of the satellite is m.



Q9. The area of the shaded region

(A) A/20 (B) A/8

(C) A/40 (D) A/4

Q10. The angular momentum of satellite per unit mass per unit area is

(A) 5×10^{-8} /s (B) 16×10^{-8} /s

(C) 2×10^{-8} /s (D) Zero

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q11.

Column–I	Column–II
(A) Kinetic energy of a particle in gravitational field is increasing	(P) work done by gravitational force should be positive
(B) Potential energy of a particle in gravitational field is increasing	(Q) work done by external force should be non zero
(C) Mechanical energy of a particle in gravitational field is increasing	(R) work done by gravitational force should be negative
	(S) can not say anything

Q12. A satellite is in a circular equatorial orbit of radius 7000 km around the Earth. If it is transferred to a circular orbit of double the radius

Column I	Column II
(A) Angular momentum	(P) Increases
(B) Area of Earth covered by satellite signal	(Q) Decreases
(C) Potential energy	(R) Becomes double
(D) Kinetic energy	(S) Becomes half

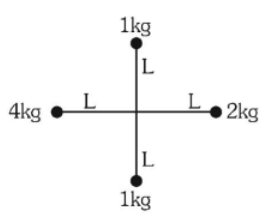
NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q13. How much below the surface does in Km the acceleration due to gravity becomes 70% of its value on the surface of earth?

SUBJECTIVE TYPE

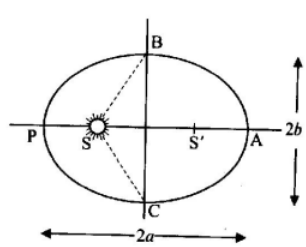
Q14. In the given figure, L = 1 metre, if total gravitational force on 4 kg mass is $\vec{F}_1$ and on 2 kg mass is $\vec{F}_2$, then find out the ratio of $\left|\frac{\vec{F}_1}{\vec{F}_2}\right|$.



Q15. A satellite of mass m is orbiting the earth in a circular orbit of radius r. It starts losing energy due to small air resistance at the rate of C J/ s. Then the time taken for the satellite to reach the earth is _____.

Q16. A comet orbits the sun in a highly elliptical orbit. Does the comet have a constant (a) linear speed, (b) angular speed, (c) angular momentum, (d) kinetic energy, (e) potential energy, (f) total energy throught out its orbit ? Neglect any mass loss of the comet when it comes very close to the Sun.

Q17. Let the speed of the planet at the perihelion P in Fig. be v_p and the Sun-planet distance SP be r_p. Relate (r_p, v_p) to the corresponding quantities at the aphelion (r_A, v_A). Will the planet take equal times to travel BAC and CPB ?



SINGLE CORRECT TYPE

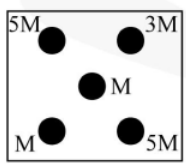
*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

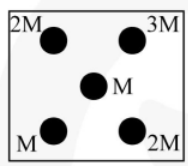
Q1. Three identical bodies (each of mass M) are placed at vertics of an equilateral triangle of arm L, keeping the triangle as such, by which angular speed the bodies should be rotated in their gravitation fields so that the triangle moves along circumferenes of circular orbit:

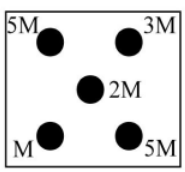
(A) $\sqrt{\frac{3GM}{L^3}}$ (B) $\sqrt{\frac{GM}{L^3}}$

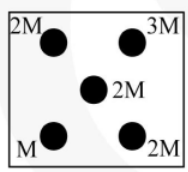
(C) $\sqrt{\frac{GM}{3L^3}}$ (D) $3\sqrt{\frac{GM}{L^3}}$

Q2. A mass is at the center of a square, with four masses at the corners as shown. Rank the choices according to the magnitude of the gravitational force on the center mass.

(A) 

(B) 

(C) 

(D) 

(A) $F_A = F_B < F_C = F_D$
(B) $F_A > F_B < F_D < F_C$
(C) $F_A = F_B > F_C = F_D$
(D) None

Q3. If the distance between the centres of earth and moon is D and mass of earth is 81 times that of moon. At what distance from the centre of earth gravitational field will be zero :-

(A) $\frac{D}{2}$ (B) $\frac{2D}{3}$ (C) $\frac{4D}{5}$ (D) $\frac{9D}{10}$

Q4. A planet whose mass and radius are both half of that of earth . Acceleration due to gravity (g) at its surface should be :-

(A) 29.4 m/sec^2 (B) 19.6 m/sec^2
(C) 9.8 m/sec^2 (D) 4.9 m/sec^2

Q5. The value of 'g' reduces to half of its value at surface of earth at a height 'h', then :–

(A) $h = R$
(B) $h = 2R$
(C) $h = (\sqrt{2}+1)R$
(D) $h = (\sqrt{2}-1)R$

Q6. If the gravitational acceleration at surface of earth is g, then increase in potential energy in lifting an object of mass m to a height equal to the radius R_e of earth will be:–

(A) $\frac{mgR_e}{2}$ (B) $2mgR_e$
(C) mgR_e (D) $\frac{mgR_e}{4}$

Q7. If the length of the day is T, the height of that TV satellite above the earth's surface which always appears stationary from earth, will be :

(A) $h = \left[\frac{4\pi^2 GM}{T^2}\right]^{1/3}$

(B) $h = \left[\frac{4\pi^2 GM}{T^2}\right]^{1/2} - R$

(C) $h = \left[\frac{GMT^2}{4\pi^2}\right]^{1/3} - R$

(D) $h = \left[\frac{GMT^2}{4\pi^2}\right]^{1/3} + R$

Q8. A geostationary satellite orbits around the Earth in a circular orbit of radius 36,000 km. Then, the time period of a spy satellite orbiting a few hundred km (600 km) above the Earth's surface (R_e= 6400 km) will approximately be :-

(A) $\frac{1}{2}$ h (B) 1h

(C) 2h (D) 4h

Q9. If a graph is plotted between T^2 and r^3 for a planet then its slope will be :-

(A) $\frac{4\pi^2}{GM}$ (B) $\frac{GM}{4\pi^2}$

(C) 4π GM (D) zero

Q10. A planet is revolving round the sun. Its distance from the sun at Apogee is r_A and that at Perigee is r_p. The mass of planet and sun is m and M respectively, v_A and v_p is the velocity of planet at Apogee and Perigee respectively and T is the time period of revolution of planet round the sun.

(a) $T^2 = \frac{\pi^2}{2Gm}(r_A + r_P)^2$

(b) $T^2 = \frac{\pi^2}{2GM}(r_A + r_P)^3$

(c) $v_A r_A = v_P r_P$

(d) $v_A < v_P$, $r_A > r_P$

(A) a, b, c (B) a, b, d

(C) b, c, d (D) all

Q11. A satellite S is moving in an elliptical orbit around the Earth. The mass of the satellite is very small compared to the mass of the Earth :-

(A) The acceleration of S always directed towards the centre of the Earth

(B) The angular momentum of S about the centre of the Earth changes in direction, but its magnitude remain constant

(C) The total mechanical energy of S varies periodically with time

(D) The linear momentum of S remains constant in magnitude

Q12. A double star system consists of two stars A and B which have time period T_A and T_B. Radius R_A and R_B and mass M_A and M_B. Choose the correct option :-

(A) If $T_A > T_B$ then $R_A > R_B$

(B) If $T_A > T_B$ then $M_A > M_B$

(C) $\left(\frac{T_A}{T_B}\right)^2 = \left(\frac{R_A}{R_B}\right)^3$

(D) $T_A = T_B$

Q13. Two planets A and B have the same material density. If the radius of A is twice that of B, then the ratio of the escape velocity $\frac{v_A}{v_B}$ is-

(A) 2 (B) $\sqrt{2}$ (C) $1/\sqrt{2}$ (D) $1/2$

Q14. If velocity of a satellite is half of escape velocity, then distance of the satellite from earth surface will be :-

(A) 6400 km (B) 12800 km

(C) $6400\sqrt{2}$ km (D) $\frac{6400}{\sqrt{2}}$ km

Q15. If the angular speed of earth is increased so much that the objects start flying from equator, then the length of the day will be nearly :-

(A) $1\frac{1}{2}$ hour (B) 8 hour

(C) 18 hour (D) 24 hour

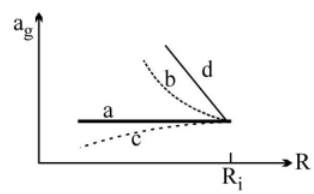
Q16. A (nonrotating) star collapses onto itself from an initial radius R_i with its mass remaining unchanged. Which curve in figure best gives the gravitational acceleration a_g on the surface of the star as a function of the radius of the star during the collapse?

(A) a

(B) b

(C) c

(D) d



MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Two masses m_1 and m_2 ($m_1 < m_2$) are released from rest from a finite distance. They start under their mutual gravitational attraction

(A) acceleration of m_1 is more than that of m_2

(B) acceleration of m_2 is more than that of m_1

(C) centre of mass of system will remain at rest in all the references frame

(D) total energy of system remains constant

Q2. At what altitude will the acceleration due to gravity be 25% of that at the earth's surface (given radius of earth is R)?

(A) R/4 (B) R

(C) 3R/8 (D) R/2

Q3. Let ω be the angular velocity of the earth's rotation about its axis. Assume that the acceleration due to gravity on the earth's surface has the same value at the equator and the poles in absence of rotation of earth. An object weighed at the equator gives the same reading as a reading taken at a depth d below earth's surface at a pole (d<<R) The value of d is

(A) $\frac{\omega^2 R^2}{g}$ (B) $\frac{\omega^2 R^2}{2g}$

(C) $\frac{2\omega^2 R^2}{g}$ (D) $\frac{\sqrt{Rg}}{g}$

Q4. A satellite is fired from the surface of the moon of mass M and radius R, with speed v_0 at 30° with the vertical. The satellite reaches a maximum distance of $\frac{5R}{2}$ from the centre of the moon. The value of v_0 is-

(A) $\sqrt{\frac{2GM}{R}}$ (B) $\sqrt{\frac{5GM}{2R}}$

(C) $\sqrt{\frac{5GM}{4R}}$ (D) $\sqrt{\frac{2GM}{37R}}$

Q5. Satellites A and B are orbiting around the earth in orbits of ratio R and 4R respectively. The ratio of their areal velocities is :

(A) 1 : 2 (B) 1 : 4

(C) 1 : 8 (D) 1 : 16

Q6. If a satellite orbits as close to the earth's surface as possible,

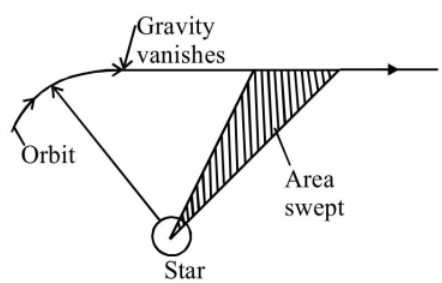
(A) Its speed is maximum

(B) Time period of its rotation is minimum

(C) The total energy of the 'earth plus satellite' system is minimum

(D) The total energy of the 'earth plus satellite'system is maximum

Q7. A planet is orbiting a star when for no apparent reason the star's gravity suddenly vanishes. After which planet moves in a straight line. Mark the **CORRECT** statement(s) :-



(A) Newton's first law is obeyed on planet after gravity vanishes

(B) Kepler's law of areas is obeyed only till the planet is in gravity of star

(C) Kepler's law of areas is obeyed even after gravity vanishes

(D) Angular momentum of planet about centre of star is conserved through out its motion

Q8. A spherical uniform planet is rotating about its axis. The velocity of a point on its equator is V. Due to the rotation of planet about its axis the acceleration due to gravity g at equator is 1/2 of g at poles. The escape velocity of a particle on the pole of planet in terms of V is

(A) $V_e = 2V$ (B) $V_e = V$

(C) $V_e = V/2$ (D) $V_e = \sqrt{3}\,V$

Q9. The escape velocity for a planet is v_e. A tunnel is dug along a diameter of the planet and a small body is dropped into it at the surface. When the body reaches the centre of the planet, its speed will be

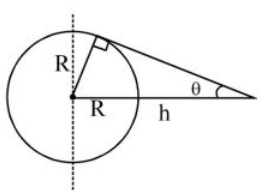
(A) v_e (B) $\frac{v_e}{\sqrt{2}}$

(C) $\frac{v_e}{2}$ (D) zero

Q10. The fractional change in the value of free–fall acceleration g for a particle when it is lifted from the surface to an elevation h (h<<R) is

(A) $\frac{h}{R}$ (B) $\frac{2h}{R}$ (C) $-\frac{2h}{R}$ (D) $-\frac{h}{R}$

Q11. A geostationary satellite is at a height h above the surface of earth. If earth radius is R.



(A) The minimum colatitude on earth upto which the satellite can be used for communication is $\sin^{-1}(R/R+h)$.

(B) The maximum latitudes on earth upto which the satellite can be used for communication is $\cos^{-1}(R/R+h)$.

(C) The area on earth escaped from this satellite is given as $2\pi R^2 (1 + \sin\theta)$

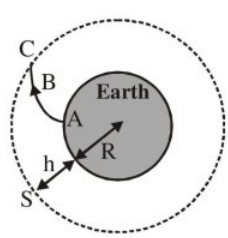
(D) The area on earth escaped from this satellite is given as $2\pi R^2 (1 + \cos\theta)$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question nos. 12 to 14

An Earth satellite is a body placed in a stable orbit about the Earth. These satellites are used for communications, meteorology, Earth measurements (gravitation and magnetic fields), resource evaluation (water, minerals), transmission of radio and TV signals, and as reference points for navigation. Although most satellites are launched from ground based stations, more recently some have been placed in orbit from one of NASA's space shuttles.There are two requirements needed to place a satellite in a stable orbit at an insertion point C. (Figure). It is first necessary to bring the satellite to that altitude and then the satellite must be given the necessary orbiting velocity. The orbiting velocity for a circular orbit, also called the insertion velocity.



Q12. What is insertion velocity for very small h (<<R)

(A) $\left(\frac{GM}{R}\right)^{1/2}\left[1-\frac{h}{2R}\right]$

(B) $\left(\frac{2GM}{R}\right)^{1/2}\left[1-\frac{h}{2R}\right]$

(C) $\left(\frac{GM}{R}\right)^{1/2}\left[1-\frac{2h}{R}\right]$

(D) $\left(\frac{GM}{2R}\right)^{1/2}\left[1-\frac{h}{2R}\right]$

Q13. If velocity at insertion point is k times escape velocity of the satellite at the insertion height h, then mark incorrect statement.

(A) For k =1 satellite moves in a circular orbit

(B) For k ≥1 satellite moves in an unbound orbit

(C) For k < 1 satellite may move in an elliptical orbit

(D) For k < 1 satellite can move in circular orbit

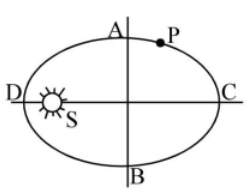
Q14. A particle is projected from surface of Earth with velocity v, such that it's total energy at height h becomes same as the total energy of a satellite, moving in circular orbit of same height. Value of v is

(A) $\sqrt{\frac{2GM}{R}}\left(1+\frac{h}{2R}\right)$ (B) $\sqrt{\frac{GM}{R}}\left(1+\frac{h}{2R}\right)$

(C) $\sqrt{\frac{2GM}{R}}\left(1-\frac{h}{2R}\right)$ (D) $\sqrt{\frac{GM}{R}}\left(1-\frac{h}{2R}\right)$

Paragraph for Question No. 15 & 16

Figure shows the orbit of a planet P around the sun S. AB and CD are the minor and major axes of the ellipse.



Q15. If t_1 is the time taken by the planet to travel along ACB and t_2 the time along BDA, then

(A) $t_1 = t_2$

(B) $t_1 > t_2$

(C) $t_1 < t_2$

(D) Nothing can be concluded

Q16. If U is the potential energy and K kinetic energy then |U| > |K| at

(A) Only D

(B) Only C

(C) both D & C

(D) neither D nor C

SUBJECTIVE TYPE

Q17. The gravitational field in a region is given by $\vec{E} = (3\hat{i} - 4\hat{j})$ N/kg. Find out the work done (in joule) in displacing a particle by 1 m, along the line 4y = 3x + 9 :

Q18. Calculate the distance from the surface of the earth at which above and below the surface acceleration due to gravity is the same.

Q19. A pair of stars rotates about a common center of mass. One of the stars has a mass M which is twice as large as the mass m of the other. Their centres are at a distance d apart, d being large compared to the size of either star.

(a) Derive an expression for the period of rotation of the stars about their common centre of mass in terms of d,m, G.

(b) Compare the angular momentum of the two stars about their common centre of mass by calculating the ratio L_m / L_M.

(c) Compare the kinetic energies of the two stars by calculating the ratio K_m/K_M.

Q20. In a particular double star system, two stars of mass 3.22×10^{30} kg each revolve about their common centre of mass, 1.12×10^{11} m away.

(i) Calculate their common period of revolution, in years.

(ii) Suppose that a meteoroid (small solid particle in space) passes through this centre of mass moving at right angles to the orbital plane of the stars. What must its speed be if it is to escape from the gravitational field of the double star?

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
A	C	C	A	C	B	A	D	D	A
11	12	13	14	15	16	17	18	19	20
B	C	C	B	C	A	B	D	C	B
21	22	23	24	25	26	27	28	29	30
A	D	A	A	C	C	D	B	B	A

Ex-1A

1. A,B **2.** D **3.** A,C,D **4.** A,C **5.** A,B,C,D

6. A **7.** A,C **8.** A,D **9.** B **10.** C

11. (A)-S,(B)-R,(C)-Q **12.** (A)-P (B)-P (C)-P,S (D)-Q,S

13. 1920 km **14.** $\frac{F_1}{F_2} = \sqrt{2}$ **15.** $t = \frac{GMm}{2C}\left(\frac{1}{R_e} - \frac{1}{r}\right)$

16. All quantities vary over an orbit except angular momentum and total energy.

17. The planet will take a longer time to traverse BAC than CPB.

Ex-2									
1	2	3	4	5	6	7	8	9	10
A	A	D	B	D	A	C	C	A	C
11	12	13	14	15	16				
A	D	A	A	A	B				

Ex-2A

1. A,D **2.** B **3.** A **4.** C **5.** A

6. A,B,C **7.** A,C,D **8.** A **9.** B **10.** C

11. A,B,C **12.** A **13.** A **14.** B **15.** B

16. C **17.** 0 **18.** $h = \frac{\sqrt{5}-1}{2}R$

19. (a) $T = \frac{2\pi d^{3/2}}{\sqrt{3Gm}}$ (b) 2 (c) 2 **20.** (i) $T = 4\pi\sqrt{\frac{r^3}{GM}}$ (ii) $v = \sqrt{\frac{4GM}{r}}$; $\left(r = \frac{d}{2}\right)$